

3(a)	1 active site, shape / tertiary structure ; 2 <u>complementary</u> to, substrate(s) / (intracellular) protein / ATP ;	2
3(b)	any three from: 1 at low(er) concentrations / AW, of, substrate / ATP, lower rate of activity with GNF-5 ; 2 as substrate concentration increases, effect of GNF-5 decreases ; 3 maximum rate of reaction / $V_{\max}$ / described, is the same ; 4 Michaelis-Menten constant / K_m, is higher with inhibitor ; ora <i>accept comparative data quotes to support answers</i> <i>for ATP concentration, must use $\mu\text{mol dm}^{-3}$ at least once</i>	3
3(c)(i)	without inhibitor, activity of T315L any one from: 1 no activity at $0.1 \mu\text{mol dm}^{-3}$ (ATP) ; 2 maximum rate of reaction / $V_{\max}$, is reached / AW, (for T315L) at a higher concentration (of ATP) ; 3 Michaelis-Menten constant / K_m, is higher (for T315L) ; ora <i>accept comparative data quotes to support answers</i> <i>for ATP concentration, must use $\mu\text{mol dm}^{-3}$ at least once</i>	1
3(c)(ii)	with inhibitor, activity of T315L any two from: 1 activity does not reach TKR, maximum rate of reaction / $V_{\max}$; 2 <i>idea that</i> drug / GNF-5, acts as a non-competitive inhibitor of T315L ; 3 drug / GNF-5, has less of an effect at low concentrations on activity (of enzyme / T315L) ; ora <i>accept comparative data quotes to support answers</i>	2